PRINTABLE STUDENT WORKSHEET / BLANK RESPONSE COPY

Strawberry DNA extraction

From fruit tissue to pale, visible fibres. Record what you predict, do and observe.

NAME	CLASS	DATE
_____	_____	_____

TEACHER-SUPERVISED PRACTICAL. Wear eye protection. Keep ethanol capped, cold and away from flames. Follow your teacher's instructions for spills, glassware and chemical handling.

01 / BEFORE THE PRACTICAL

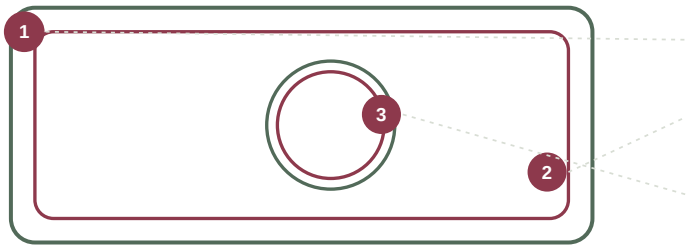
Before the practical

Think first; return to your predictions after the experiment.

1.1 Cultivated garden strawberries are octoploid (8n). In your own words, what does 8n tell you about chromosome sets? Does chromosome-set count alone tell you the total DNA mass in a cell? Explain.

1.2 Use the word bank to label the numbered barriers in the simplified cell sketch.

WORD BANK: cell wall / cell membrane / nuclear envelope



1. _____
2. _____
3. _____

SIMPLIFIED CELL SKETCH - NOT TO SCALE

1.3 Before adding the cold liquid, where do you predict the pale fibres will appear? Describe what you expect to see.

02 / METHOD AND OBSERVATION RECORD

Method and observation record

Use the lesson's five moves. Keep the liquid layers separate and write only what you observe.

BUFFER MIX	EQUIPMENT AND MATERIALS
90 cm3 water 10 cm3 detergent 3 g salt **Use 10 cm3 of prepared buffer on the mashed fruit.**	One hulled strawberry; sealable bag; gauze-lined funnel; clean tube; clean glass rod; goggles; ice-cold ethanol.

FIVE MOVES

MOVE	ACTION	DONE
01	Place one hulled strawberry in a sealed bag. Mash gently for 2 minutes; avoid whipping in air or making foam.	[ ]
02	Measure 90 cm3 water and 10 cm3 detergent. Stir in 3 g salt until dissolved. Add 10 cm3 of this buffer to the mashed fruit.	[ ]
03	Pour slowly through gauze in a funnel into a clean tube. Collect the liquid; leave most visible pulp on the gauze.	[ ]
04	Measure 5 cm3 filtrate and an equal 5 cm3 volume of ice-cold ethanol. Run ethanol slowly down the tube wall; do not shake or stir.	[ ]
05	Touch a clean glass rod to the boundary between the liquids. Turn slowly in one direction to spool any pale, stringy material.	[ ]

OBSERVATION LOG

STAGE	WHAT I SEE
After mashing and adding buffer	
After filtration: liquid and retained pulp	
At the ethanol-filtrate boundary	
After spooling: colour, texture and location	

Look for pale, stringy fibres at the boundary only after cold ethanol meets the filtrate. Keep ethanol capped and away from ignition sources.

SAFETY CHECK	BEFORE YOU PACK UP
[ ] Eye protection on [ ] Ethanol capped and away from flames [ ] Glassware intact; spills or breakage reported using class procedure	One observation I trust: _____ _____ One question I still have: _____ _____

03 / EXPLAIN AND EVALUATE

Explain and evaluate

Use evidence from your own sample. A visible result is an observation, not proof of purity.

WHAT DOES EACH STEP DO? Write an explanation in your own words.

STEP OR MATERIAL	MY EXPLANATION
Mashing	
Detergent	
Salt	
Filtration	
Ice-cold ethanol	
Spooling with a glass rod	

FILTRATION CHECK / Tick one destination for each item.

MATERIAL	RETAINED BY GAUZE	IN THE FILTRATE
Coarse pulp and large cell-wall fragments	[ ]	[ ]
Unmashed strawberry tissue	[ ]	[ ]
Soluble DNA	[ ]	[ ]
Water	[ ]	[ ]
Salt ions and detergent molecules	[ ]	[ ]

3.1 Why is salt added to the buffer? Include DNA's charge in your explanation.

3.2 Why use ice-cold ethanol and add it gently without mixing the layers?

3.3 What can you conclude from the fibres you observed? What can appearance alone not prove?